Business Statistics

Order-based Measures of Dispersion



Some graphics and datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Measures of Dispersion
Overview

- The dispersion of a variable has two main characteristics:
  - Spread - the distance observations are from the center
  - Shape - how observations are distributed around the center.
- Order-based dispersions
  - Range
  - IQR
- Value-based dispersions
  - Mean Absolute (Percentage) Deviation – MAD / MAPD
  - Variance, Standard Deviation, Coefficient of Variation

Dispersion of a variable

A variable's values are "dispersed"



Dispersion of a Variable
Why do observations of a variable differ?

- Measurement variability:
  - Differences in how the observation was measured
- Natural variability :
  - Members of the population or sample are different
- Induced variability:
  - In experiments differences arise due to treatments between the control and experimental groups
- Sample variability:
  - Different samples include different members of the population

Dispersion of a Variable
A variable takes on multiple values

- As observations have different values, measures of **dispersion** (spread) describe how far those observations are from the center
- Greater dispersion indicates greater variability among the observations
- The observations' dispersion affects:
  - How likely future observations are to be close to the center
  - The variability of calculations based on those observations
- Different measures give different information, and we consider multiple measures to get the whole picture
- One group of these measures are based on having the observations ordered from lowest to highest value

Dispersion of a variable

Order-based measures



Order-based measures
Calculations are based on positions of values

- Order-based measures of dispersion describe the spread of observations based on their relative position in the sorted data
- Like order-based measures of location, the observations must be sorted by value from low to high
- Calculations for dispersion measures are based on observation values
- The observation values to use are determined by the value positions and the specific measure being calculated

Order-based measures
Range captures the overall spread of observations

- The **range** of values that a variable can take is a direct measure of how far the variable values are "spread out"

Range = maximum value – minimum value

- The range gives the overall spread and is directly impacted by outliers
- Here are the hours studied by students in preparation for an exam:

	Hours studied					Range of hours
Class section 001	4.8	4.9	5	5.1	5.2	0.4
Class section 002	1	3	5	7	9	8
Class section 003	1	3	5	7	40	39

- Section 1 has a much tighter range – they all could have studied together
- Sections 2 and 3 have mostly the same values, but their ranges differ
- Section 3's "over-achiever" makes the 2 sections look wildly different

Order-based measures
Quartiles require observations to be ordered

- The pth percentile divides a variable's ordered observations into 2 groups:
  - Approximately $p\%$ below and $(100 - p)\%$ above the pth percentile
- Recall the **median** is the 50th percentile, dividing the n observations in half
  - When n is odd the median is the observation at position $i = \frac{n+1}{2}$
  - When n is even the median is the average of the observations bracketing position $i = \frac{n+1}{2}$
- **Quartiles** divide the n observations into 4 equal groups of observations
  - Q1 is the 25th percentile – 25% of observations below this value
  - Q2 is the 50th percentile – 50% of observations below this value (median)
  - Q3 is the 75th percentile – 75% of observations below this value

Order-based measures
Quartiles are the medians of the halves of data

- To calculate the quartiles, first find the median, and then:
  - Q1 (25th) percentile will be the median of the lower half of observations
  - Q3 (75th) percentile will be the median of the upper half of observations

- The same approach for calculating the median applies
  - Let n be the number of observations in the lower and upper halves of observations
  - n is odd: median is the observation at position $i = \frac{n+1}{2}$
  - n is even: median is the average of 2 observations bracketing position $i = \frac{n+1}{2}$

Order-based measures
Calculating quartiles by hand: n is odd

Index i	1	2	3	4	5	6	7	8	9
Data	10	20	30	40	50	60	70	80	90

$$Q1 = 25 \qquad Q2 = 50 \qquad Q3 = 75$$

- $n = 9 \Rightarrow i = \frac{9+1}{2} = 5$ so Q2 is the value at position 5

- Lower half has indices 1-4 so $\boldsymbol{n = 4}$
  - $\boldsymbol{n = 4} \Rightarrow \boldsymbol{i = \frac{4+1}{2} = 2.5}$
  - Q1 = average of the 2nd and 3rd values of the lower range

$$\boldsymbol{Q1 = \frac{20 + 30}{2} = 25}$$

- Upper half has indices 6-9 so $\boldsymbol{n = 4}$
  - $\boldsymbol{n = 4} \Rightarrow \boldsymbol{i = \frac{4+1}{2} = 2.5}$
  - Q3 = average of the 2nd and 3rd values of the upper range

$$\boldsymbol{Q3 = \frac{70 + 80}{2} = 75}$$

Order-based measures
Calculating quartiles by hand: n is even

Index i	1	2	3	4	5	6	7	8	9	10
Data	10	20	30	40	50	60	70	80	90	100

$\boldsymbol{Q1 = 30}$ $\boldsymbol{Q2 = 55}$ $\boldsymbol{Q3 = 80}$

- $n = 10 \Rightarrow i = \frac{10+1}{2} = 5.5$
  - Q2 average of the 5th and 6th observations $\boldsymbol{Q2 = \frac{50 + 60}{2} = 55}$

- Lower half has indices 1-5 so $\boldsymbol{n = 5}$
  - $\boldsymbol{n = 5} \Rightarrow \boldsymbol{i = \frac{5+1}{2} = 3}$
  - Q1 at 3rd position in lower range

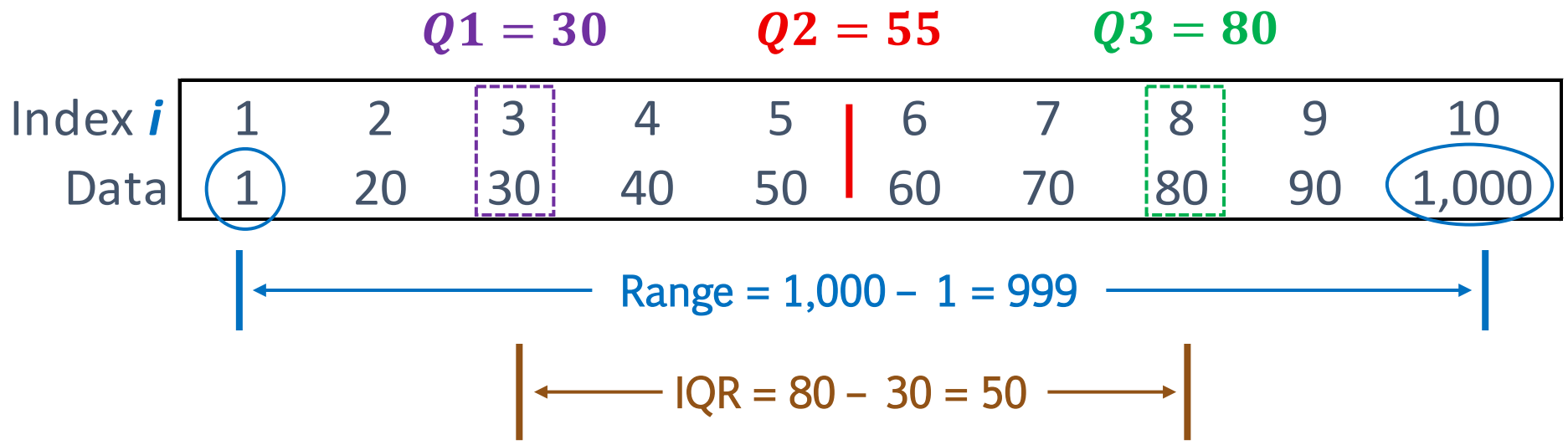
$\boldsymbol{Q1 = 30}$

- Upper half has indices 6-10 so $\boldsymbol{n = 5}$
  - $\boldsymbol{n = 5} \Rightarrow \boldsymbol{i = \frac{5+1}{2} = 3}$
  - Q3 at 3rd position in upper range

$\boldsymbol{Q3 = 80}$

Order-based measures
The interquartile range (IQR)

- The **range** measures the spread of all values, and is influenced by outliers
- The Interquartile Range (IQR) measures the spread of the middle 50% of observations, which are the "typical" observations around the center
- The IQR is the range of values between the 25th (Q1) and 75th (Q3) percentile



Order-based measures
5 Number Summary

- Order-based measures can be summarized with 5 numbers
  - Minimum, Q1, Q2 (median), Q3, maximum

Index *i*	1	2	3	4	5	6	7	8	9	10
Data	1	20	30	40	50	60	70	80	90	1,000

Min	Q1	Q2	Q3	Max
1	30	55	80	1000

- 5 Number Summary Excel Formulas:

  Min: =min(<data range>)
  Q1: =quartile.inc(<data range>, 1)
  Q2: =quartile.inc(<data range>, 2)
  Q3: =quartile.inc(<data range>, 3)
  Max: =max(<data range>)

- Excel has 2 quartile functions:
  - They end in either .inc or .exc:
  - "=quartile.inc" or "=quartile.exc"
- This decides whether to include or exclude the median in the upper and lower halves
- Stick with the .inc version. That is also how R and Python calculate quartiles

Order-based measures
Outliers

- Outliers are atypical observations, and are based on the IQR
  - Low outliers: observations where $x_i \leq Q1 - (1.5 * IQR)$
  - High outliers: observations where $x_i \geq Q3 + (1.5 * IQR)$

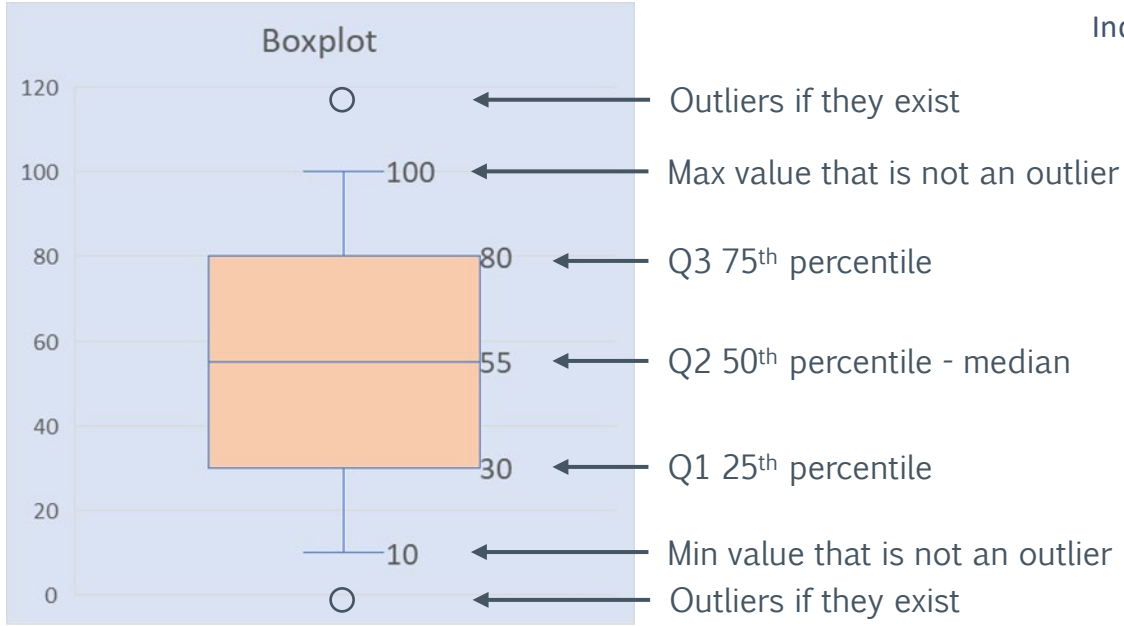
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Data	1	20	30	40	50	60	70	80	90	1,000

Min	Q1	Q2	Q3	Max
1	30	55	80	1000

- $IQR = 80 - 30 = 50$
- $1.5 * IQR = 1.5 * 50 = 75$
- Outliers:
  - $x_i \leq 30 - 75 = -45$
  - $x_i \geq 80 + 75 = 155$
  - $x_{10} = 1,000 \geq 155$
  - Observation 10 is an outlier

Order-based measures
Visualizing the 5 number summary

- The 5 number summary is visualized with a boxplot



Index *i*	1	2	3	4	5	6	7	8	9	10
Data	10	20	30	40	50	60	70	80	90	100

Min	Q1	Q2	Q3	Max
10	30	55	80	100

- The above summary and boxplot are based on the hand-calculated Q1 and Q3.
- Note: in practice quartiles are calculated using software (Excel, R) which often DO NOT return the same quartiles as have been calculated by hand using the median of the lower and upper ranges.